Solution
Clearly, if ε is real and $\hat{G}$ is Hermitian, the operator $e^{i\varepsilon\hat{G}}$ would be unitary. Using the property $[F(\hat{A})]^\dagger = F^*(\hat{A}^\dagger)$, we see that

$$(e^{i\varepsilon\hat{G}})^\dagger = e^{-i\varepsilon\hat{G}} = (e^{i\varepsilon\hat{G}})^{-1}, \tag{2.125}$$

that is, $\hat{U}^\dagger = \hat{U}^{-1}$.

2.4.8 Eigenvalues and Eigenvectors of an Operator

Having studied the properties of operators and states, we are now ready to discuss how to find the eigenvalues and eigenvectors of an operator.

A state vector $|\psi\rangle$ is said to be an *eigenvector* (also called an eigenket or eigenstate) of an operator $\hat{A}$ if the application of $\hat{A}$ to $|\psi\rangle$ gives

$$\hat{A}\,|\psi\rangle = a\,|\psi\rangle, \tag{2.126}$$

where a is a complex number, called an *eigenvalue* of $\hat{A}$. This equation is known as the *eigenvalue equation*, or *eigenvalue problem*, of the operator $\hat{A}$. Its solutions yield the eigenvalues and eigenvectors of $\hat{A}$. In Section 2.5.3 we will see how to solve the eigenvalue problem in a discrete basis.

A simple example is the eigenvalue problem for the unity operator $\hat{I}$:

$$\hat{I}\,|\psi\rangle = |\psi\rangle. \tag{2.127}$$

This means that all vectors are eigenvectors of $\hat{I}$ with one eigenvalue, 1. Note that

$$\hat{A}\,|\psi\rangle = a\,|\psi\rangle \implies \hat{A}^n\,|\psi\rangle = a^n\,|\psi\rangle \quad \text{and} \quad F(\hat{A})\,|\psi\rangle = F(a)\,|\psi\rangle. \tag{2.128}$$

For instance, we have

$$\hat{A}\,|\psi\rangle = a\,|\psi\rangle \quad \implies \quad e^{i\hat{A}}\,|\psi\rangle = e^{ia}\,|\psi\rangle. \tag{2.129}$$

Example 2.11 (Eigenvalues of the inverse of an operator)
Show that if $\hat{A}^{-1}$ exists, the eigenvalues of $\hat{A}^{-1}$ are just the inverses of those of $\hat{A}$.

Solution
Since $\hat{A}^{-1}\hat{A} = \hat{I}$ we have on the one hand

$$\hat{A}^{-1}\hat{A}\,|\psi\rangle = |\psi\rangle, \tag{2.130}$$

and on the other hand

$$\hat{A}^{-1}\hat{A}\,|\psi\rangle = \hat{A}^{-1}(\hat{A}\,|\psi\rangle) = a\hat{A}^{-1}\,|\psi\rangle. \tag{2.131}$$

Combining the previous two equations, we obtain

$$a\hat{A}^{-1}\,|\psi\rangle = |\psi\rangle, \tag{2.132}$$

hence

$$\hat{A}^{-1} \mid \psi\rangle = \frac{1}{a} \mid \psi\rangle. \tag{2.133}$$

This means that $\mid \psi\rangle$ is also an eigenvector of $\hat{A}^{-1}$ with eigenvalue $1/a$. That is, if $\hat{A}^{-1}$ exists, then

$$\boxed{\hat{A} \mid \psi\rangle = a \mid \psi\rangle \quad \Longrightarrow \quad \hat{A}^{-1} \mid \psi\rangle = \frac{1}{a} \mid \psi\rangle.} \tag{2.134}$$

Some useful theorems pertaining to the eigenvalue problem

Theorem 2.1 *For a Hermitian operator, all of its eigenvalues are real and the eigenvectors corresponding to different eigenvalues are orthogonal.*

$$\text{If } \hat{A}^{\dagger} = \hat{A}, \quad \hat{A} \mid \phi_n\rangle = a_n \mid \phi_n\rangle \quad \Longrightarrow \quad a_n = \text{real number, and } \langle \phi_m \mid \phi_n\rangle = \delta_{mn}. \tag{2.135}$$

Proof of Theorem 2.1
Note that

$$\hat{A} \mid \phi_n\rangle = a_n \mid \phi_n\rangle \quad \Longrightarrow \quad \langle \phi_m \mid \hat{A} \mid \phi_n\rangle = a_n \langle \phi_m \mid \phi_n\rangle, \tag{2.136}$$

and

$$\langle \phi_m \mid \hat{A}^{\dagger} = a_m^* \langle \phi_m \mid \quad \Longrightarrow \quad \langle \phi_m \mid \hat{A}^{\dagger} \mid \phi_n\rangle = a_m^* \langle \phi_m \mid \phi_n\rangle. \tag{2.137}$$

Subtracting (2.137) from (2.136) and using the fact that $\hat{A}$ is Hermitian, $\hat{A} = \hat{A}^{\dagger}$, we have

$$(a_n - a_m^*) \langle \phi_m \mid \phi_n\rangle = 0. \tag{2.138}$$

Two cases must be considered separately:

- Case $m = n$: since $\langle \phi_n \mid \phi_n\rangle > 0$, we must have $a_n = a_n^*$; hence the eigenvalues a_n must be real.
- Case $m \neq n$: since in general $a_n \neq a_m^*$, we must have $\langle \phi_m \mid \phi_n\rangle = 0$; that is, $\mid \phi_m\rangle$ and $\mid \phi_n\rangle$ must be orthogonal.

Theorem 2.2 *The eigenstates of a Hermitian operator define a complete set of mutually orthonormal basis states. The operator is diagonal in this eigenbasis with its diagonal elements equal to the eigenvalues. This basis set is unique if the operator has no degenerate eigenvalues and not unique (in fact it is infinite) if there is any degeneracy.*

Theorem 2.3 *If two Hermitian operators, $\hat{A}$ and $\hat{B}$, commute and if $\hat{A}$ has no degenerate eigenvalue, then each eigenvector of $\hat{A}$ is also an eigenvector of $\hat{B}$. In addition, we can construct a common orthonormal basis that is made of the joint eigenvectors of $\hat{A}$ and $\hat{B}$.*

Proof of Theorem 2.3
Since $\hat{A}$ is Hermitian with no degenerate eigenvalue, to each eigenvalue of $\hat{A}$ there corresponds only one eigenvector. Consider the equation

$$\hat{A} \mid \phi_n\rangle = a_n \mid \phi_n\rangle. \tag{2.139}$$

Since $\hat{A}$ commutes with $\hat{B}$ we can write

$$\hat{B}\hat{A} \mid \phi_n\rangle = \hat{A}\hat{B} \mid \phi_n\rangle \quad \text{or} \quad \hat{A}(\hat{B} \mid \phi_n\rangle) = a_n(\hat{B} \mid \phi_n\rangle); \tag{2.140}$$

that is, $(\hat{B} \mid \phi_n\rangle)$ is an eigenvector of $\hat{A}$ with eigenvalue a_n. But since this eigenvector is unique (apart from an arbitrary phase constant), the ket $\mid \phi_n\rangle$ must also be an eigenvector of $\hat{B}$:

$$\hat{B} \mid \phi_n\rangle = b_n \mid \phi_n\rangle. \tag{2.141}$$

Since each eigenvector of $\hat{A}$ is also an eigenvector of $\hat{B}$ (and vice versa), both of these operators must have a common basis. This basis is unique; it is made of the joint eigenvectors of $\hat{A}$ and $\hat{B}$. This theorem also holds for any number of mutually commuting Hermitian operators.

Now, if a_n is a degenerate eigenvalue, we can only say that $\hat{B} \mid \phi_n\rangle$ is an eigenvector of $\hat{A}$ with eigenvalue a_n; $\mid \phi_n\rangle$ is not necessarily an eigenvector of $\hat{B}$. If one of the operators is degenerate, there exist an infinite number of orthonormal basis sets that are common to these two operators; that is, the joint basis does exist and it is not unique.

Theorem 2.4 *The eigenvalues of an anti-Hermitian operator are either purely imaginary or equal to zero.*

Theorem 2.5 *The eigenvalues of a unitary operator are complex numbers of moduli equal to one; the eigenvectors of a unitary operator that has no degenerate eigenvalues are mutually orthogonal.*

Proof of Theorem 2.5
Let $\mid \phi_n\rangle$ and $\mid \phi_m\rangle$ be eigenvectors to the unitary operator $\hat{U}$ with eigenvalues a_n and a_m, respectively. We can write

$$(\langle\phi_m \mid \hat{U}^\dagger)(\hat{U} \mid \phi_n\rangle) = a_m^* a_n \langle\phi_m \mid \phi_n\rangle. \tag{2.142}$$

Since $\hat{U}^\dagger\hat{U} = \hat{I}$ this equation can be rewritten as

$$(a_m^* a_n - 1)\langle\phi_m \mid \phi_n\rangle = 0, \tag{2.143}$$

which in turn leads to the following two cases:

- Case $n = m$: since $\langle\phi_n \mid \phi_n\rangle > 0$ then $a_n^* a_n = \mid a_n \mid^2 = 1$, and hence $\mid a_n \mid = 1$.
- Case $n \neq m$: the only possibility for this case is that $\mid \phi_m\rangle$ and $\mid \phi_n\rangle$ are orthogonal, $\langle\phi_m \mid \phi_n\rangle = 0$.

2.4.9 Infinitesimal and Finite Unitary Transformations

We want to study here how quantities such as kets, bras, operators, and scalars transform under unitary transformations. A unitary transformation is the application of a unitary operator $\hat{U}$ to one of these quantities.

2.4.9.1 Unitary Transformations

Kets $\mid \psi\rangle$ and bras $\langle \psi \mid$ transform as follows:

$$\mid \psi'\rangle = \hat{U} \mid \psi\rangle, \qquad \langle \psi' \mid = \langle \psi \mid \hat{U}^\dagger. \tag{2.144}$$

Let us now find out how operators transform under unitary transformations. Since the transform of $\hat{A} \mid \psi\rangle = \mid \phi\rangle$ is $\hat{A}' \mid \psi'\rangle = \mid \phi'\rangle$, we can rewrite $\hat{A}' \mid \psi'\rangle = \mid \phi'\rangle$ as $\hat{A}'\hat{U} \mid \psi\rangle = \hat{U} \mid \phi\rangle = \hat{U}\hat{A} \mid \psi\rangle$ which, in turn, leads to $\hat{A}'\hat{U} = \hat{U}\hat{A}$. Multiplying both sides of $\hat{A}'\hat{U} = \hat{U}\hat{A}$ by $\hat{U}^\dagger$ and since $\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \hat{I}$, we have

$$\hat{A}' = \hat{U}\hat{A}\hat{U}^\dagger, \qquad \hat{A} = \hat{U}^\dagger\hat{A}'\hat{U}. \tag{2.145}$$

The results reached in (2.144) and (2.145) may be summarized as follows:

$$\boxed{\mid \psi'\rangle = \hat{U} \mid \psi\rangle, \qquad \langle \psi' \mid = \langle \psi \mid \hat{U}^\dagger, \qquad \hat{A}' = \hat{U}\hat{A}\hat{U}^\dagger,} \tag{2.146}$$

$$\boxed{\mid \psi\rangle = \hat{U}^\dagger \mid \psi'\rangle, \qquad \langle \psi \mid = \langle \psi' \mid \hat{U}, \qquad \hat{A} = \hat{U}^\dagger\hat{A}'\hat{U}.} \tag{2.147}$$

Properties of unitary transformations

- If an operator $\hat{A}$ is Hermitian, its transformed $\hat{A}'$ is also Hermitian, since

$$\hat{A}'^\dagger = (\hat{U}\hat{A}\hat{U}^\dagger)^\dagger = \hat{U}\hat{A}^\dagger\hat{U}^\dagger = \hat{U}\hat{A}\hat{U}^\dagger = \hat{A}'. \tag{2.148}$$

- The eigenvalues of $\hat{A}$ and those of its transformed $\hat{A}'$ are the same:

$$\hat{A} \mid \psi_n\rangle = a_n \mid \psi_n\rangle \quad \Longrightarrow \quad \hat{A}' \mid \psi_n'\rangle = a_n \mid \psi_n'\rangle, \tag{2.149}$$

since

$$\begin{aligned} \hat{A}' \mid \psi_n'\rangle &= (\hat{U}\hat{A}\hat{U}^\dagger)(\hat{U} \mid \psi_n\rangle) = \hat{U}\hat{A}(\hat{U}^\dagger\hat{U}) \mid \psi_n\rangle \\ &= \hat{U}\hat{A} \mid \psi_n\rangle = a_n(\hat{U} \mid \psi_n\rangle) = a_n \mid \psi_n'\rangle. \end{aligned} \tag{2.150}$$

- Commutators that are equal to (complex) numbers remain unchanged under unitary transformations, since the transformation of $[\hat{A}, \hat{B}] = a$, where a is a complex number, is given by

$$\begin{aligned} [\hat{A}', \hat{B}'] &= [\hat{U}\hat{A}\hat{U}^\dagger, \hat{U}\hat{B}\hat{U}^\dagger] = (\hat{U}\hat{A}\hat{U}^\dagger)(\hat{U}\hat{B}\hat{U}^\dagger) - (\hat{U}\hat{B}\hat{U}^\dagger)(\hat{U}\hat{A}\hat{U}^\dagger) \\ &= \hat{U}[\hat{A}, \hat{B}]\hat{U}^\dagger = \hat{U}a\hat{U}^\dagger = a\hat{U}\hat{U}^\dagger = a \\ &= [\hat{A}, \hat{B}]. \end{aligned} \tag{2.151}$$

- We can also verify the following general relations:

$$\hat{A} = \beta\hat{B} + \gamma\hat{C} \quad \Longrightarrow \quad \hat{A}' = \beta\hat{B}' + \gamma\hat{C}', \tag{2.152}$$

$$\hat{A} = \alpha\hat{B}\hat{C}\hat{D} \quad \Longrightarrow \quad \hat{A}' = \alpha\hat{B}'\hat{C}'\hat{D}', \tag{2.153}$$

where $\hat{A}'$, $\hat{B}'$, $\hat{C}'$, and $\hat{D}'$ are the transforms of $\hat{A}$, $\hat{B}$, $\hat{C}$, and $\hat{D}$, respectively.

- Since the result (2.151) is valid for any complex number, we can state that complex numbers, such as $\langle \psi \mid \hat{A} \mid \chi \rangle$, remain unchanged under unitary transformations, since

$$\langle \psi' \mid \hat{A}' \mid \chi' \rangle = (\langle \psi \mid \hat{U}^\dagger)(\hat{U}\hat{A}\hat{U}^\dagger)(\hat{U} \mid \chi\rangle) = \langle \psi \mid (\hat{U}^\dagger\hat{U})\hat{A}(\hat{U}^\dagger U) \mid \chi \rangle = \langle \psi \mid \hat{A} \mid \chi \rangle. \tag{2.154}$$

  Taking $\hat{A} = \hat{I}$ we see that scalar products of the type

$$\langle \psi' \mid \chi' \rangle \; = \; \langle \psi \mid \chi \rangle \tag{2.155}$$

  are invariant under unitary transformations; notably, the norm of a state vector is conserved:

$$\langle \psi' \mid \psi' \rangle \; = \; \langle \psi \mid \psi \rangle. \tag{2.156}$$

- We can also verify that $\left(\hat{U}\hat{A}\hat{U}^\dagger\right)^n = \hat{U}\hat{A}^n\hat{U}^\dagger$ since

$$\begin{aligned} \left(\hat{U}\hat{A}\hat{U}^\dagger\right)^n &= \left(\hat{U}\hat{A}\hat{U}^\dagger\right)\left(\hat{U}\hat{A}\hat{U}^\dagger\right)\cdots\left(\hat{U}\hat{A}\hat{U}^\dagger\right) = \hat{U}\hat{A}(\hat{U}^\dagger\hat{U})\hat{A}(\hat{U}^\dagger\hat{U})\cdots(\hat{U}^\dagger\hat{U})\hat{A}\hat{U}^\dagger \\ &= \hat{U}\hat{A}^n\hat{U}^\dagger. \end{aligned} \tag{2.157}$$

- We can generalize the previous result to obtain the transformation of any operator function $f(\hat{A})$:

$$\hat{U}f(\hat{A})\hat{U}^\dagger = f(\hat{U}\hat{A}\hat{U}^\dagger) = f(\hat{A}'), \tag{2.158}$$

  or more generally

$$\hat{U}f(\hat{A},\hat{B},\hat{C},\cdots)\hat{U}^\dagger = f(\hat{U}\hat{A}\hat{U}^\dagger,\hat{U}\hat{B}\hat{U}^\dagger,\hat{U}\hat{C}\hat{U}^\dagger,\cdots) = f(\hat{A}',\hat{B}',\hat{C}',\cdots). \tag{2.159}$$

A unitary transformation does not change the physics of a system; it merely transforms one description of the system to another physically equivalent description.

In what follows we want to consider two types of unitary transformations: infinitesimal transformations and finite transformations.

2.4.9.2 Infinitesimal Unitary Transformations

Consider an operator $\hat{U}$ which depends on an infinitesimally small real parameter ε and which varies only slightly from the unity operator $\hat{I}$:

$$\hat{U}_\varepsilon(\hat{G}) = \hat{I} + i\varepsilon\hat{G}, \tag{2.160}$$

where $\hat{G}$ is called the *generator* of the infinitesimal transformation. Clearly, $\hat{U}_\varepsilon$ is a unitary transformation only when the parameter ε is real and $\hat{G}$ is Hermitian, since

$$\hat{U}_\varepsilon\hat{U}_\varepsilon^\dagger = (\hat{I} + i\varepsilon\hat{G})(\hat{I} - i\varepsilon\hat{G}^\dagger) \simeq \hat{I} + i\varepsilon(\hat{G} - \hat{G}^\dagger) = \hat{I}, \tag{2.161}$$

where we have neglected the quadratic terms in ε.

The transformation of a state vector $\mid \psi\rangle$ is

$$\mid \psi'\rangle \; = \; (\hat{I} + i\varepsilon\hat{G}) \mid \psi\rangle \; = \mid \psi\rangle + \delta \mid \psi\rangle, \tag{2.162}$$

where

$$\delta \mid \psi\rangle = i\varepsilon\hat{G} \mid \psi\rangle. \tag{2.163}$$

The transformation of an operator $\hat{A}$ is given by

$$\boxed{\hat{A}' = (\hat{I} + i\varepsilon\hat{G})\hat{A}(\hat{I} - i\varepsilon\hat{G}) \simeq \hat{A} + i\varepsilon[\hat{G}, \hat{A}].} \tag{2.164}$$

If $\hat{G}$ commutes with $\hat{A}$, the unitary transformation will leave $\hat{A}$ unchanged, $\hat{A}' = \hat{A}$:

$$[\hat{G}, \hat{A}] = 0 \quad \Longrightarrow \quad \hat{A}' = (\hat{I} + i\varepsilon\hat{G})\hat{A}(\hat{I} - i\varepsilon\hat{G}) = \hat{A}. \tag{2.165}$$

2.4.9.3 Finite Unitary Transformations

We can construct a *finite* unitary transformation from (2.160) by performing a succession of infinitesimal transformations in steps of ε; the application of a series of successive unitary transformations is equivalent to the application of a single unitary transformation. Denoting $\varepsilon = \alpha/N$, where N is an integer and α is a finite parameter, we can apply the same unitary transformation N times; in the limit $N \to +\infty$ we obtain

$$\hat{U}_\alpha(\hat{G}) = \lim_{N\to\infty} \prod_{k=1}^{N} \left(1 + i\frac{\alpha}{N}\hat{G}\right) = \lim_{N\to+\infty} \left(1 + i\frac{\alpha}{N}\hat{G}\right)^N = e^{i\alpha\hat{G}}, \tag{2.166}$$

where $\hat{G}$ is now the generator of the finite transformation and α is its parameter.

As shown in (2.125), $\hat{U}$ is unitary only when the parameter α is real and $\hat{G}$ is Hermitian, since

$$(e^{i\alpha\hat{G}})^\dagger = e^{-i\alpha\hat{G}} = (e^{i\alpha\hat{G}})^{-1}. \tag{2.167}$$

Using the commutation relation (2.117), we can write the transformation $\hat{A}'$ of an operator $\hat{A}$ as follows:

$$\boxed{e^{i\alpha\hat{G}}\hat{A}e^{-i\alpha\hat{G}} = \hat{A} + i\alpha[\hat{G}, \hat{A}] + \frac{(i\alpha)^2}{2!}\left[\hat{G}, [\hat{G}, \hat{A}]\right] + \frac{(i\alpha)^3}{3!}\left[\hat{G}, [\hat{G}, [\hat{G}, \hat{A}]]\right] + \cdots.} \tag{2.168}$$

If $\hat{G}$ commutes with $\hat{A}$, the unitary transformation will leave $\hat{A}$ unchanged, $\hat{A}' = \hat{A}$:

$$[\hat{G}, \hat{A}] = 0 \Longrightarrow \hat{A}' = e^{i\alpha\hat{G}}\hat{A}e^{-i\alpha\hat{G}} = \hat{A}. \tag{2.169}$$

In Chapter 3, we will consider some important applications of infinitesimal unitary transformations to study time translations, space translations, space rotations, and conservation laws.

2.5 Representation in Discrete Bases

By analogy with the expansion of Euclidean space vectors in terms of the basis vectors, we need to express any ket $\mid \psi\rangle$ of the Hilbert space in terms of a complete set of mutually orthonormal base kets. State vectors are then represented by their components in this basis.